Version Control and Collaboration

Basics

Why Version Control? The Problem

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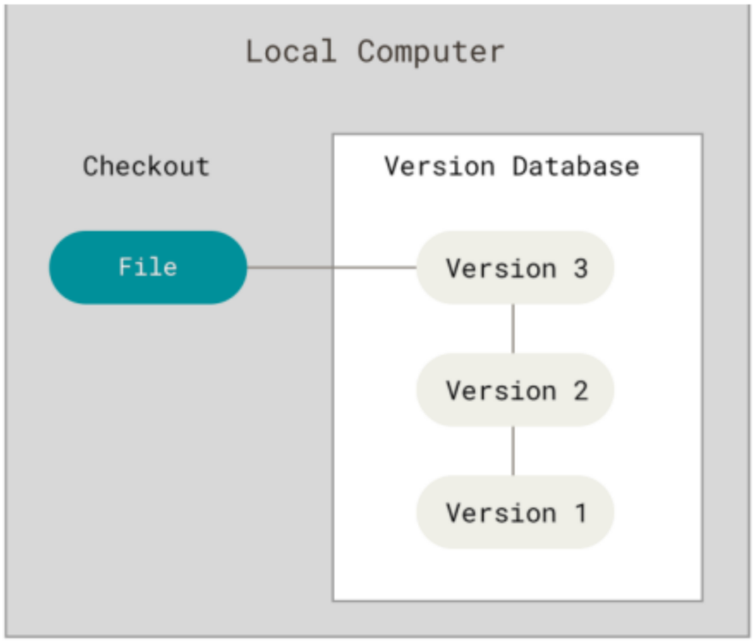
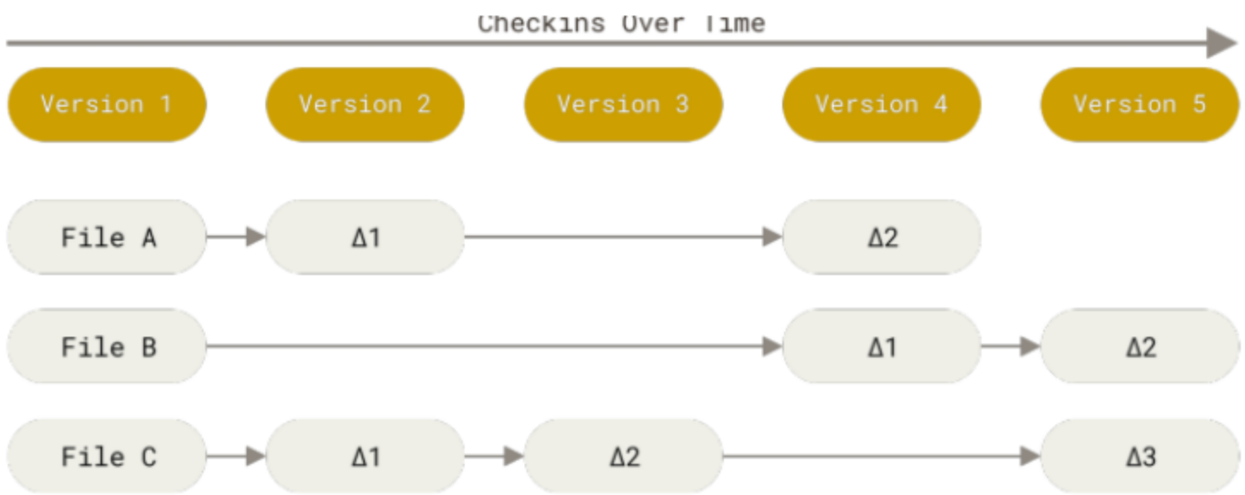
- No clear history
- Difficult collaboration
- Easy to lose work

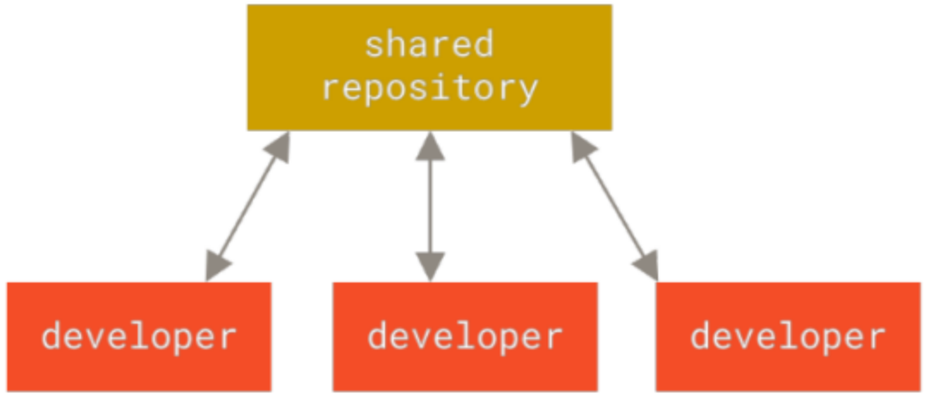
Local Version Control System



*R*CS - Idea: Keep patch sets that can used to recreate a file



Centralized Version Control (SVN)



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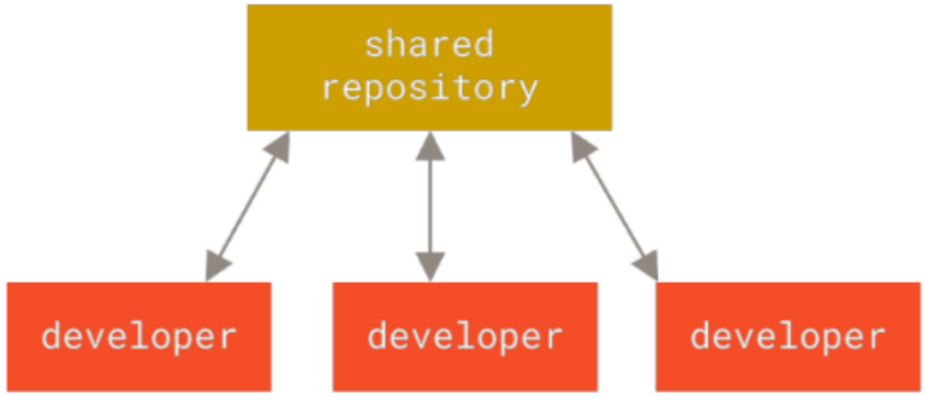

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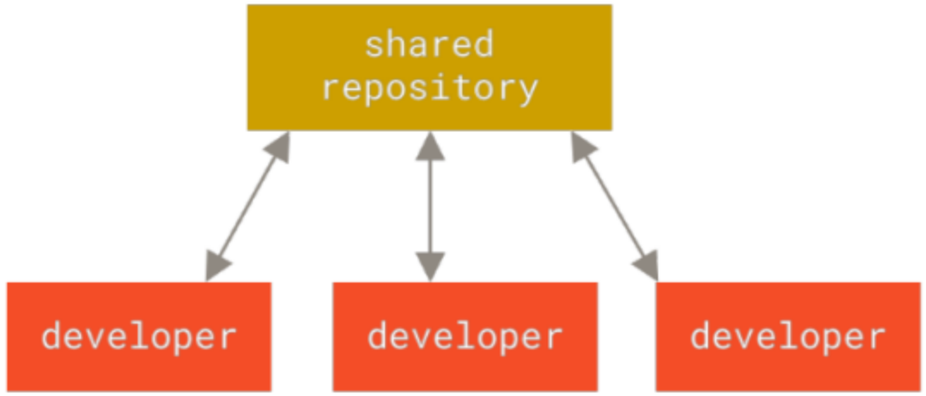

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- Clients checkout working copies

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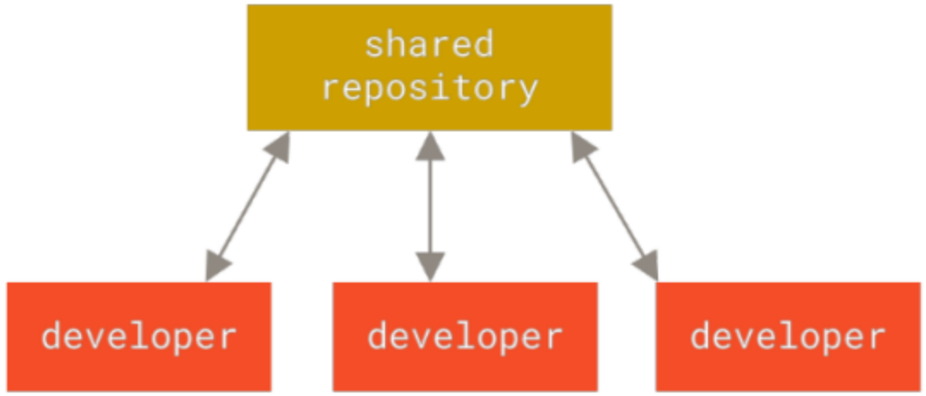

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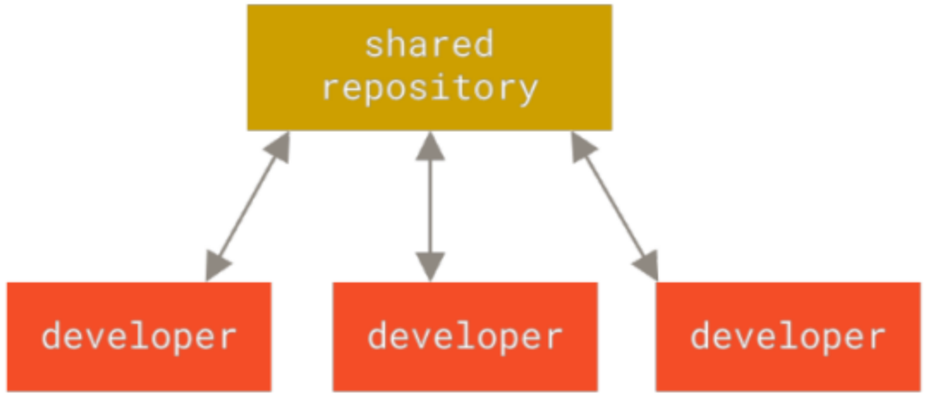

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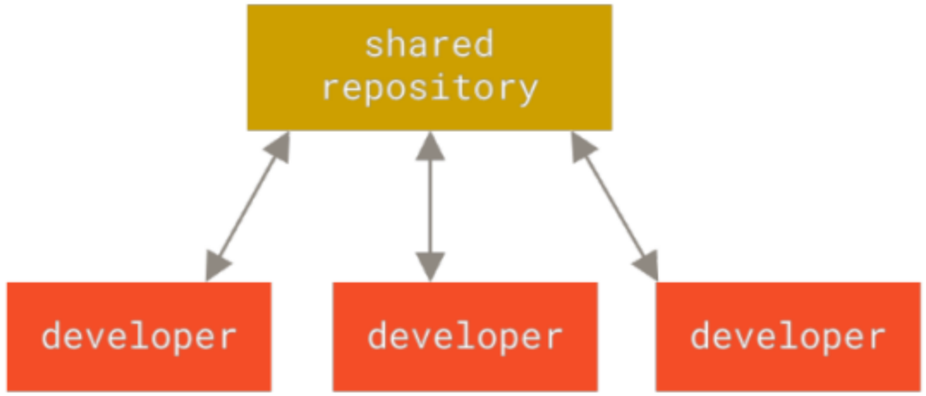

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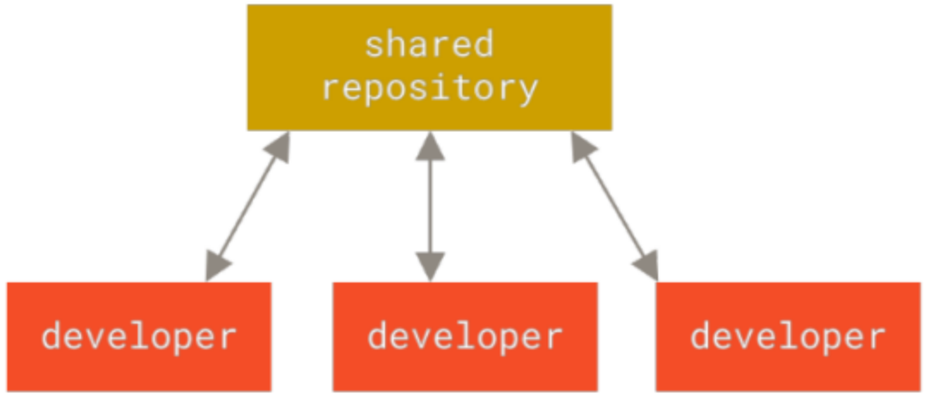

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Key Limitation: Single point of failure

Why Git?

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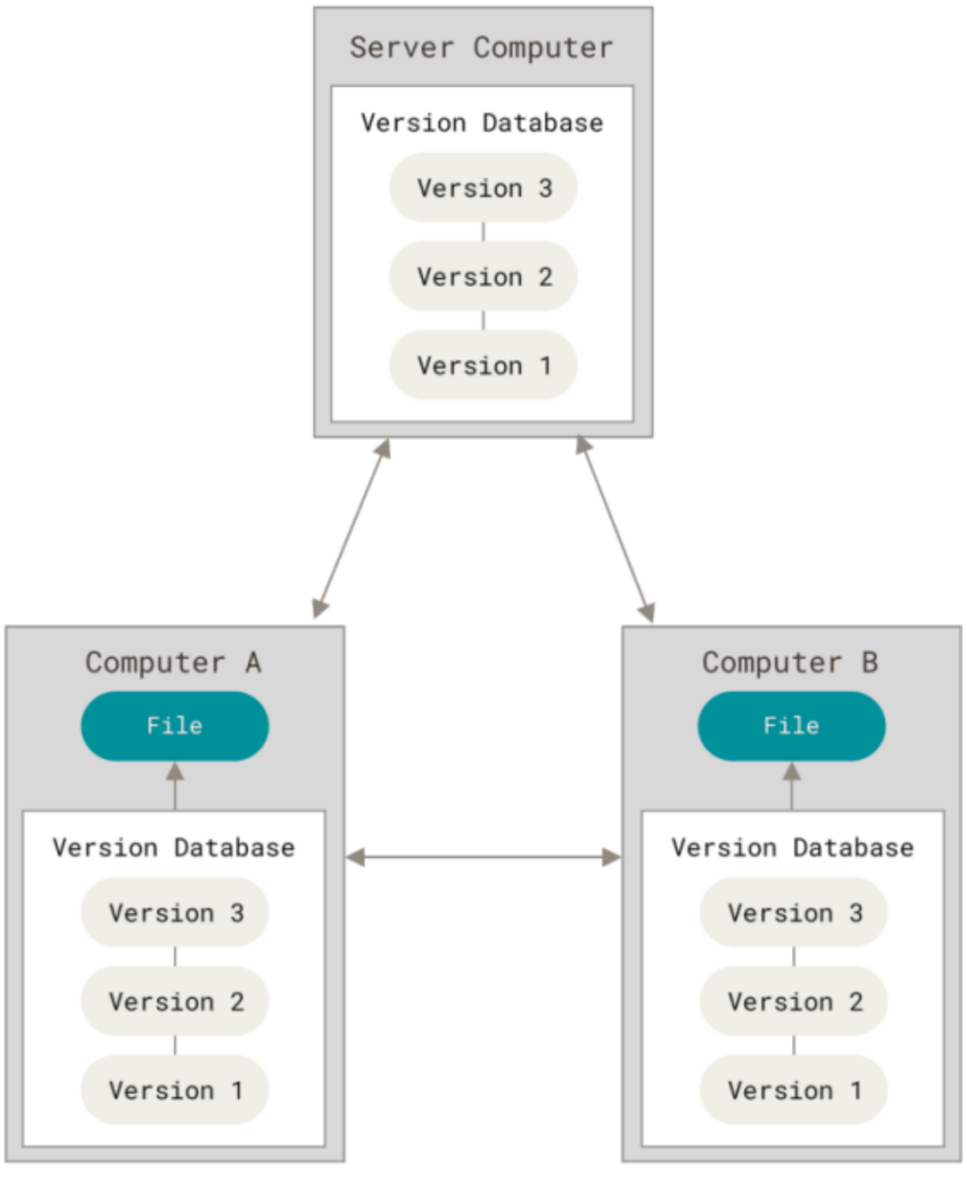

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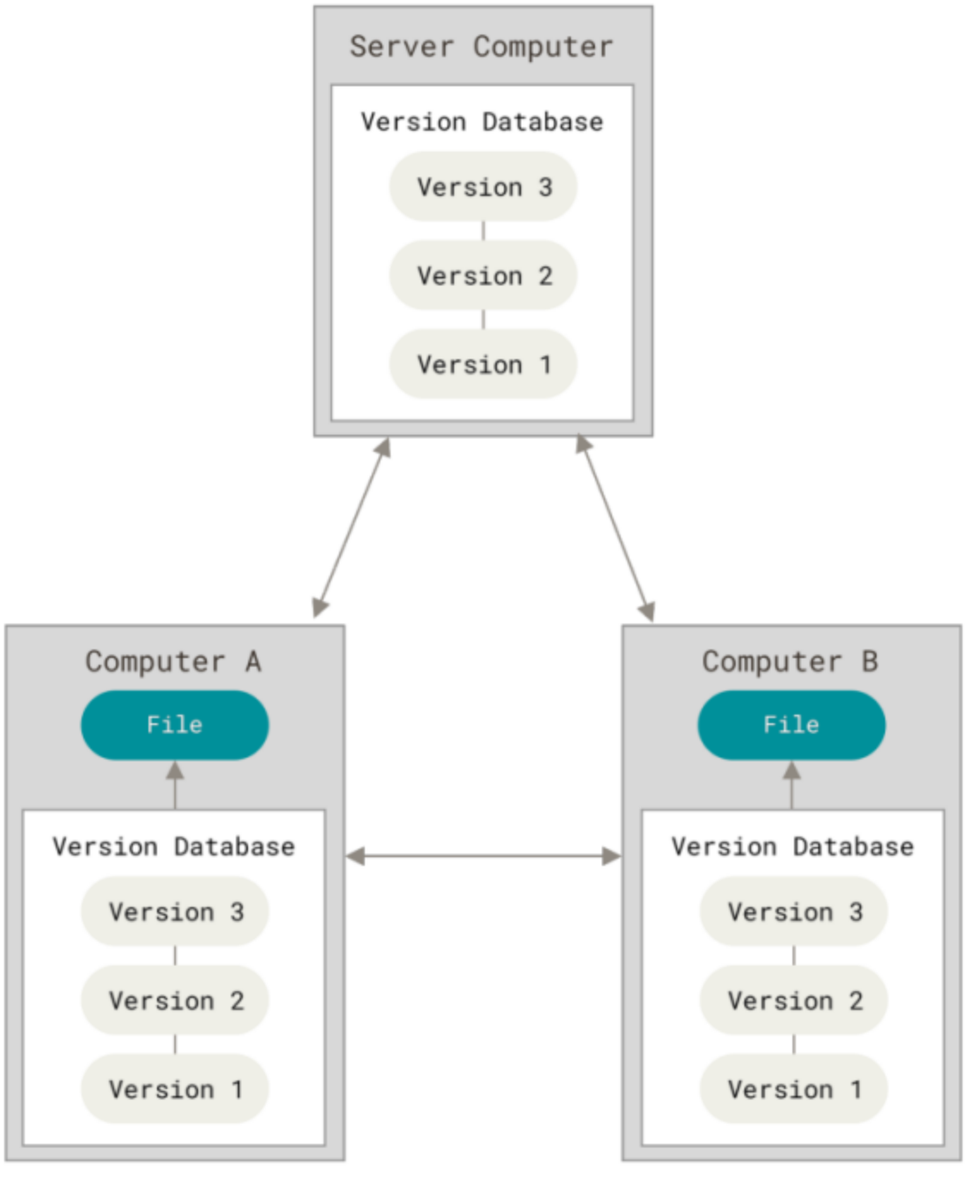
- Distributed version control
- Every clone has full history



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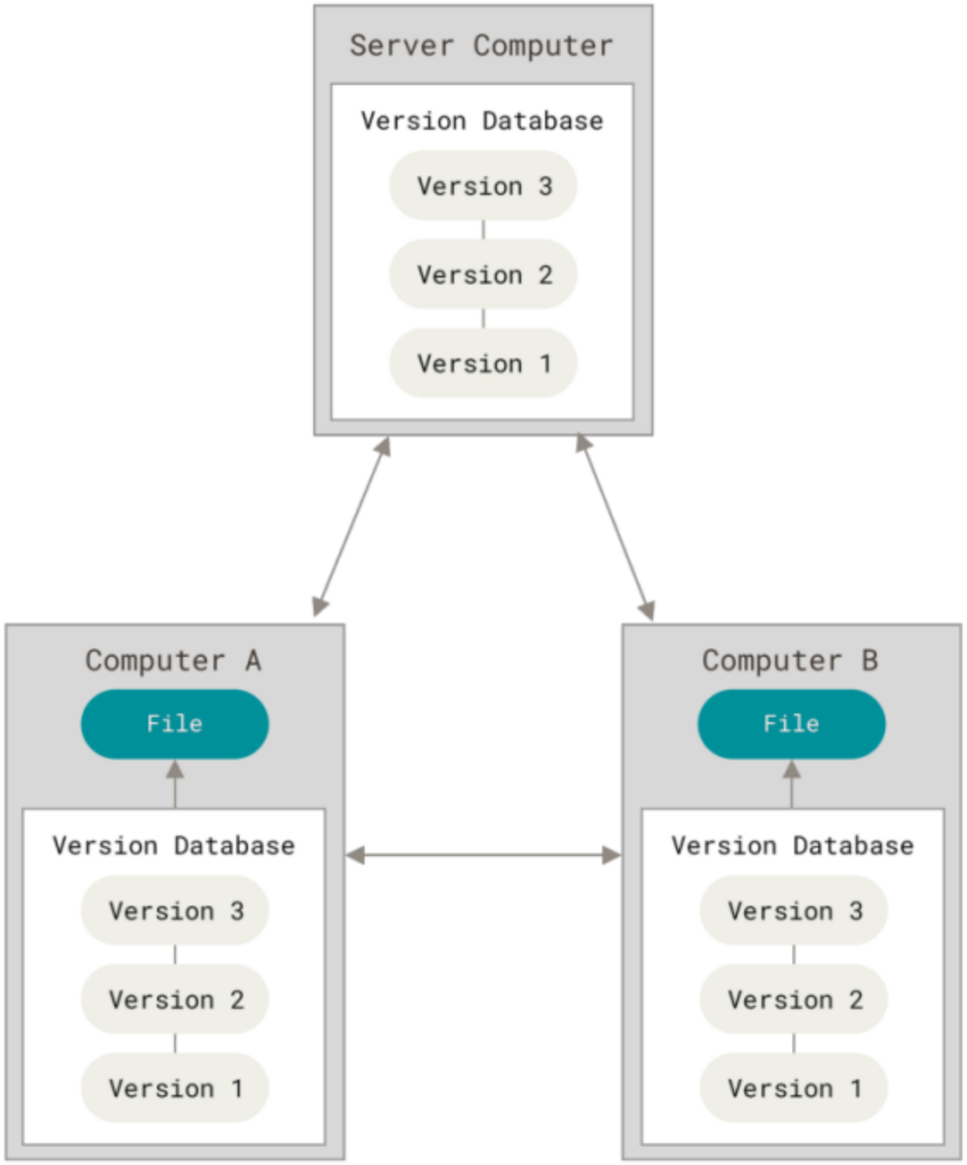

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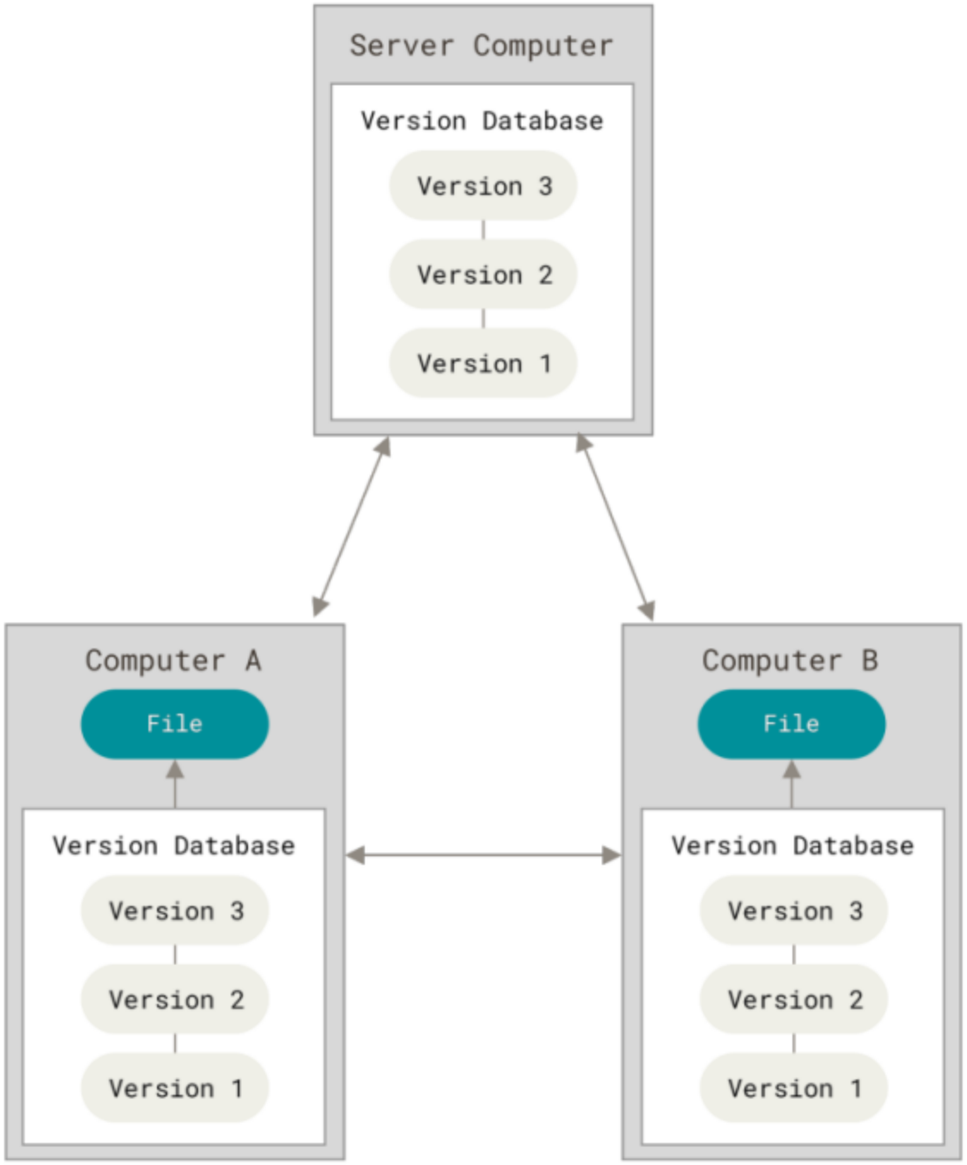

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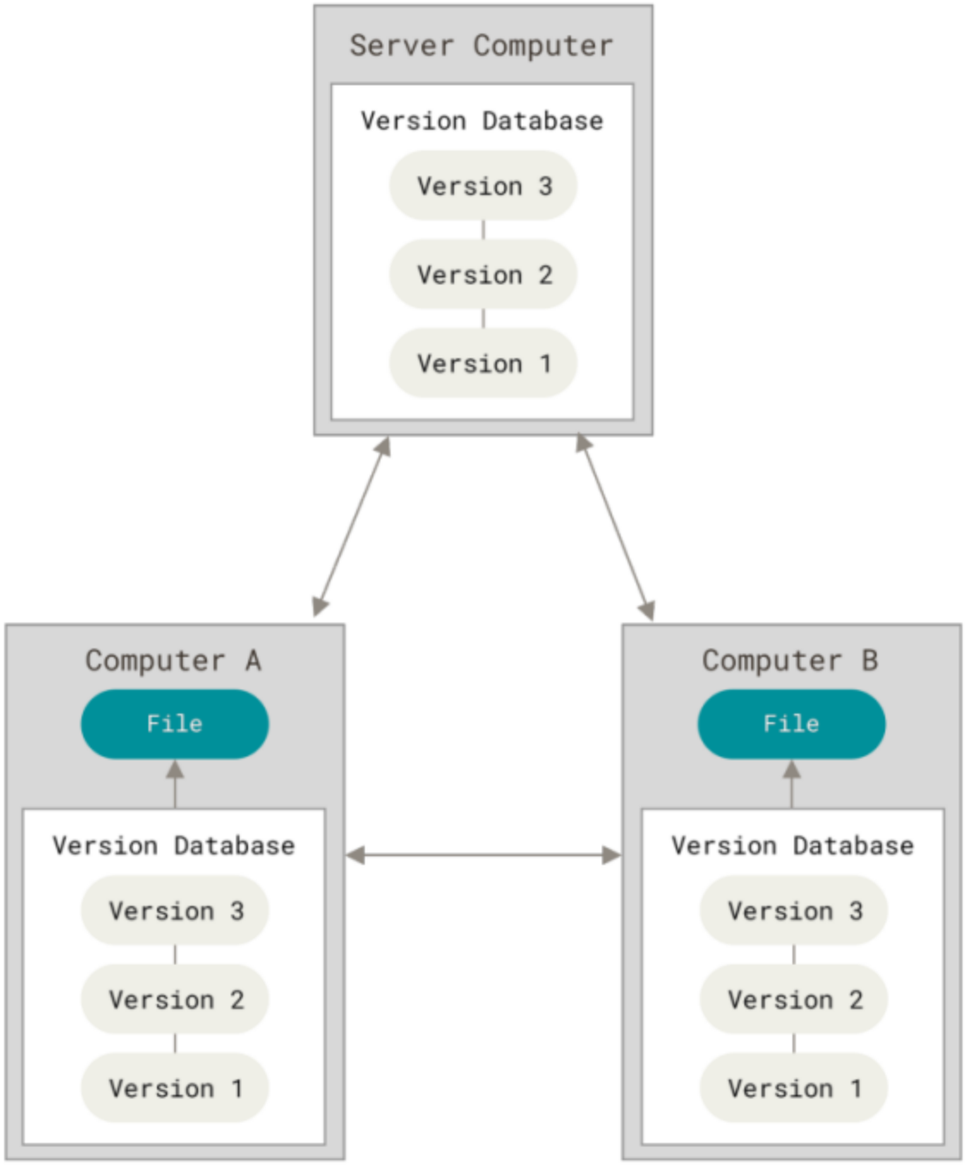

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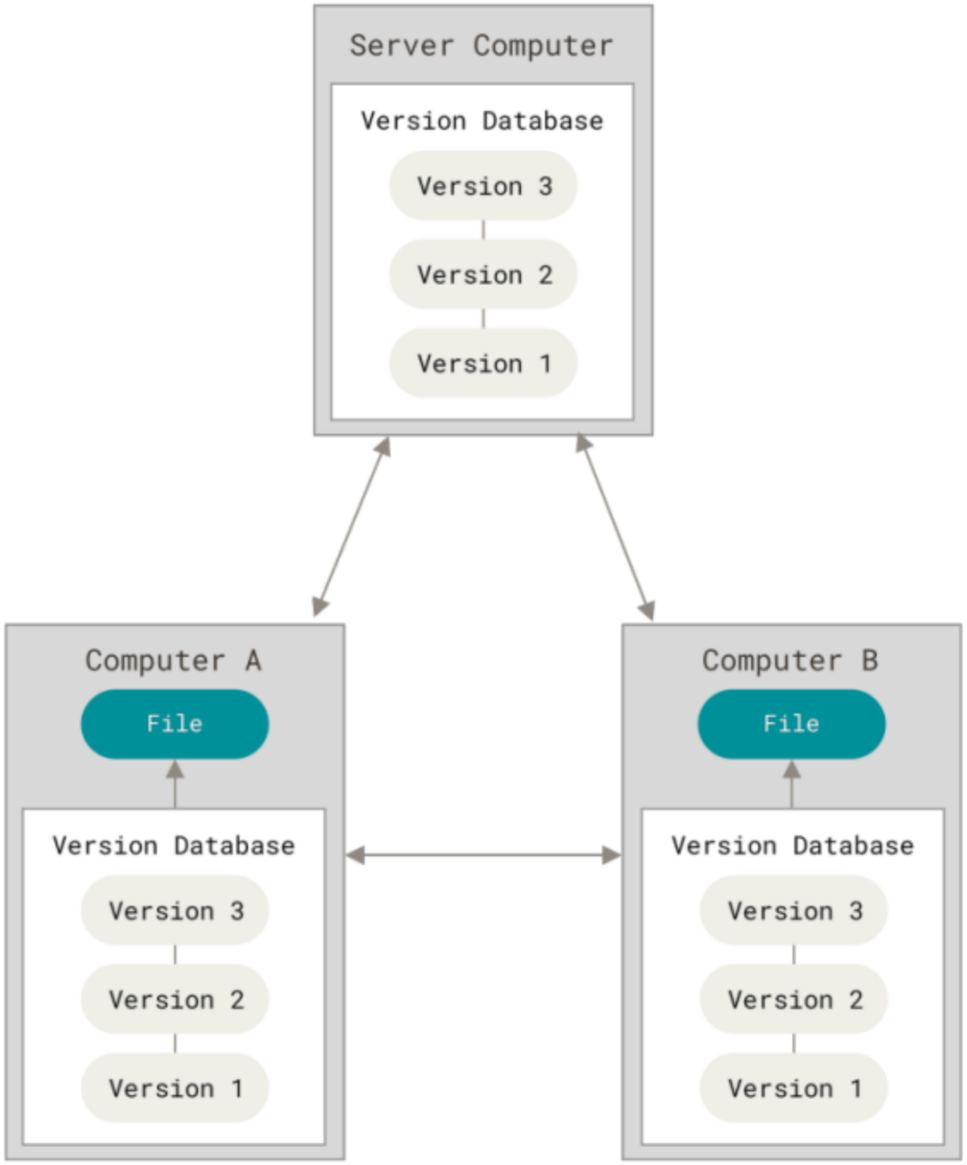
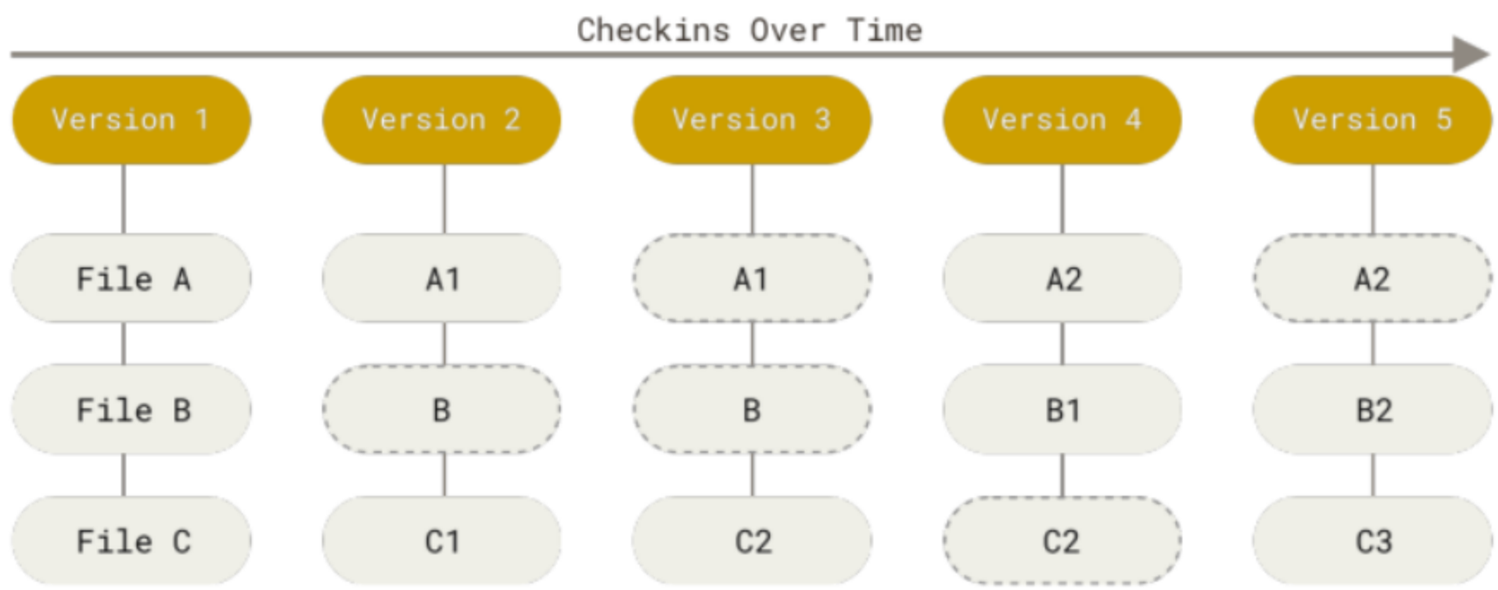


Key Idea: Collaboration without central dependency

Git's Mental Model



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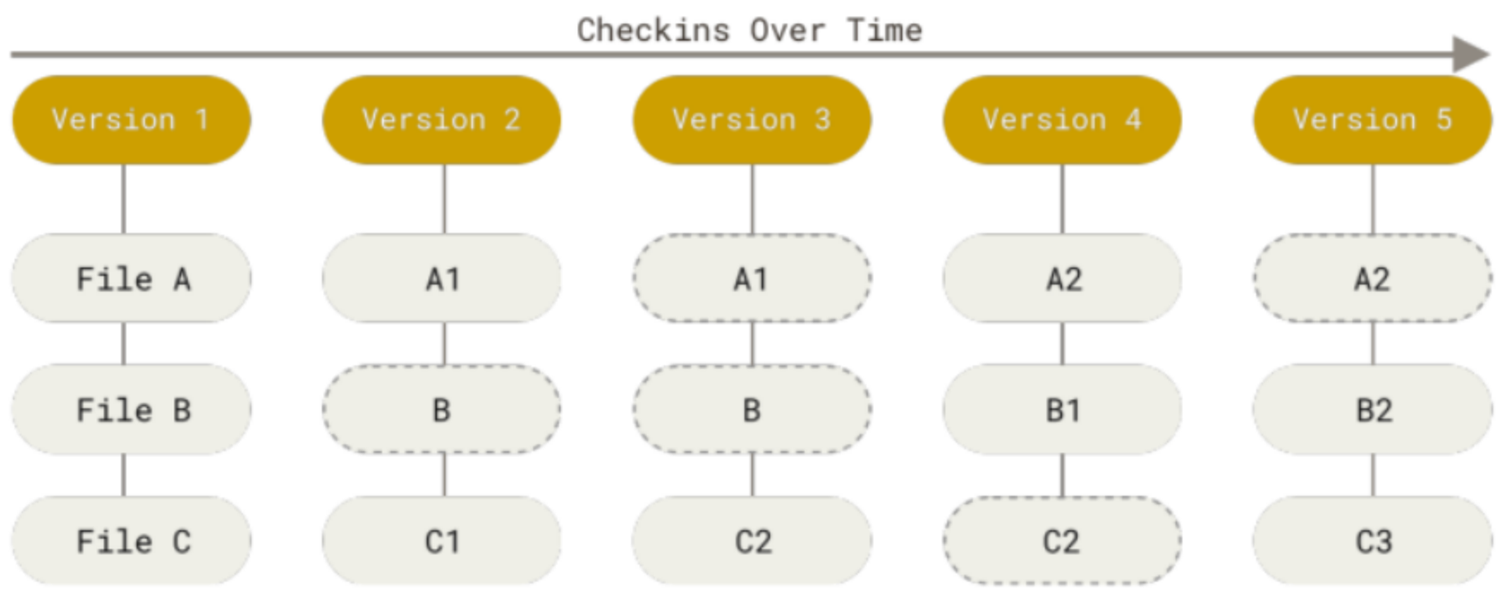

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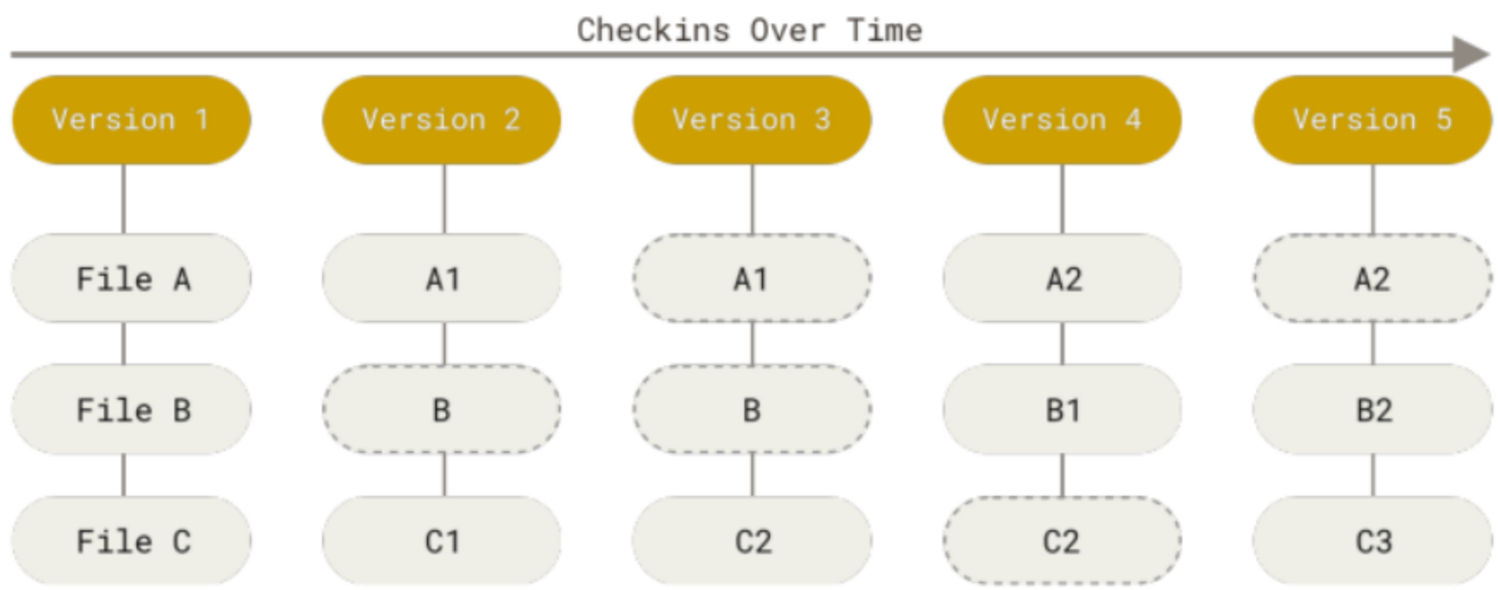

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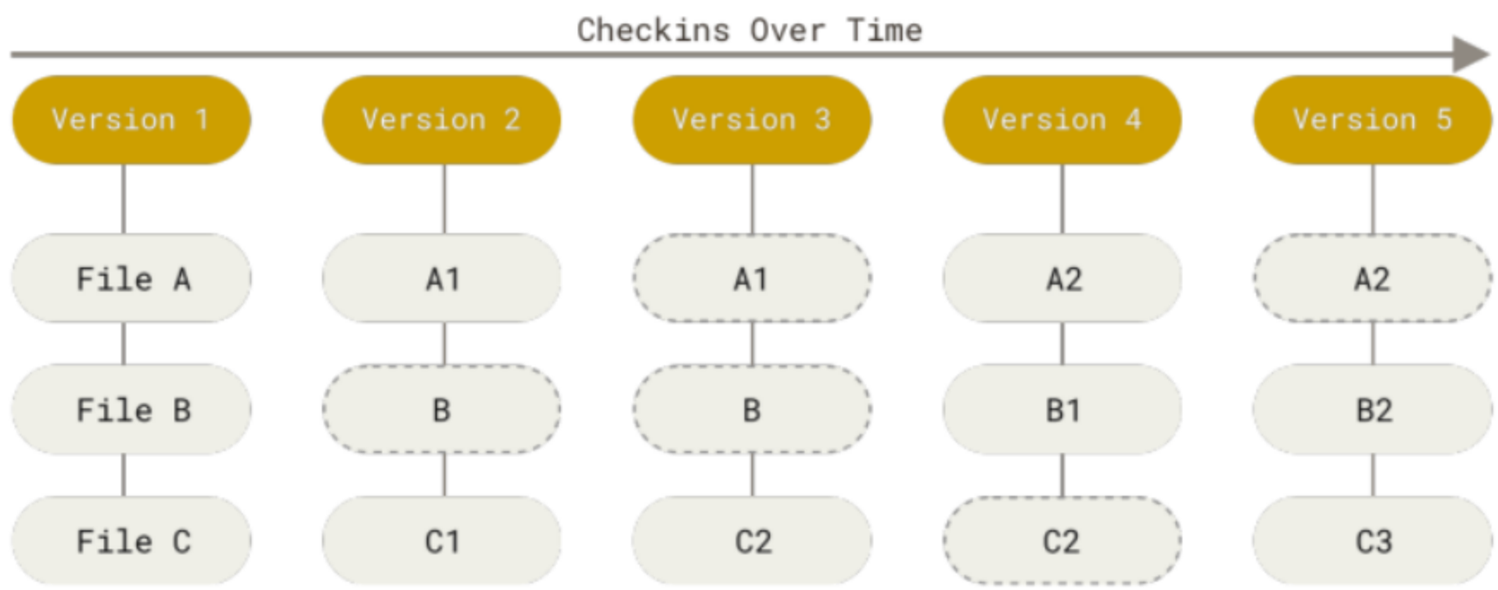

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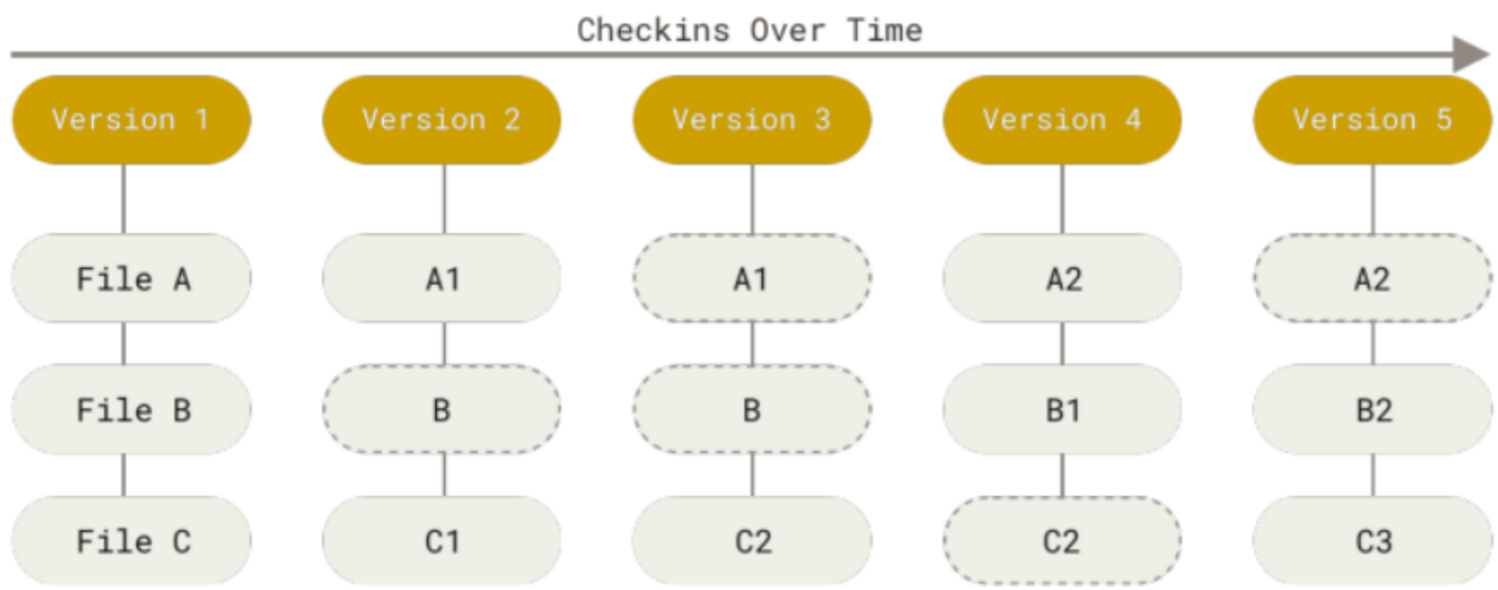

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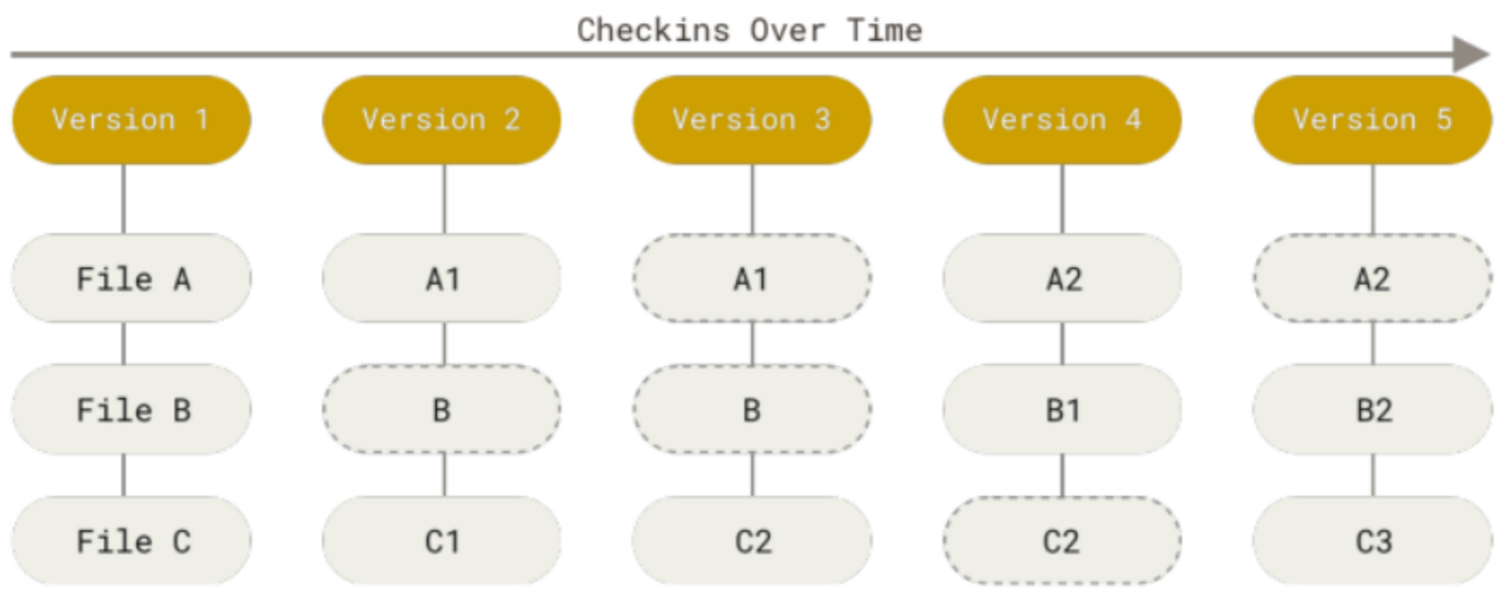

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- Each commit points to:
  - A snapshot of files
  - Parent commit(s)
- History forms a DAG (Directed Acyclic Graph)
- Nearly all operations are local

Git Configuration and Initializing a Repository

Commands

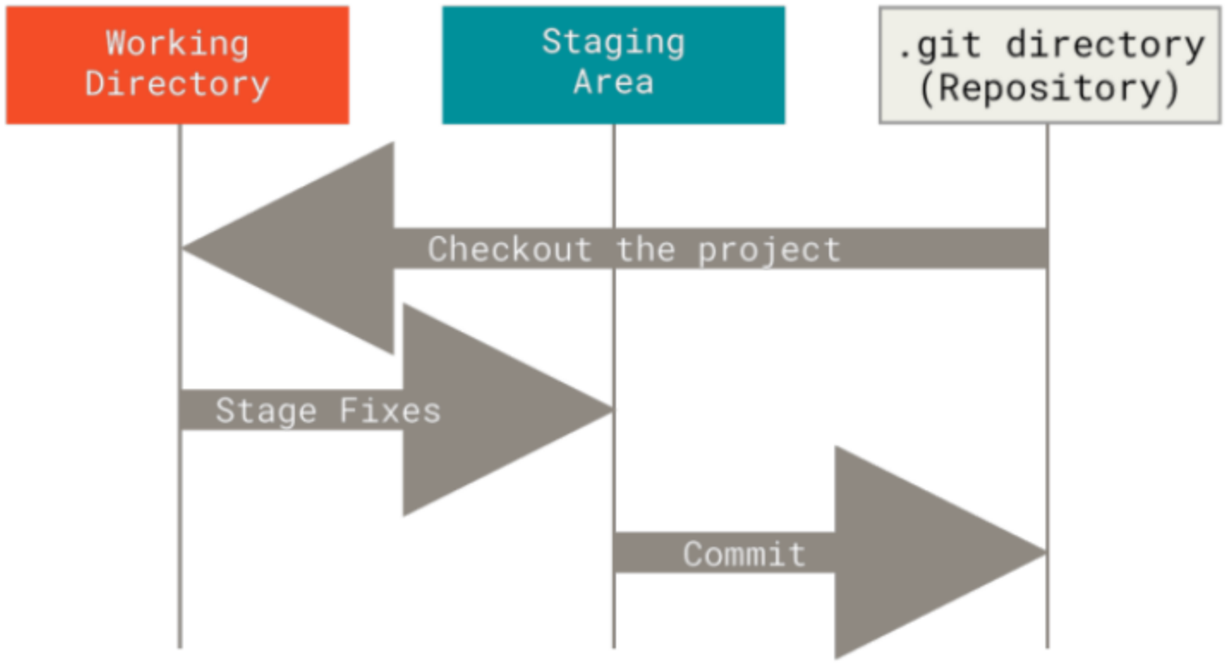
```
git config --global user.name "Kalpesh Kapoor"
git config --global user.email kalpesh.kapoor@iiserpune.ac.in
mkdir dspractice
cd dspractice
git init
```

- git global configuration in home directory `~/.gitconfig`

```
[user]
 email = kalpesh.kapoor@iiserpune.ac.in
 name  = Kalpesh
...
```

- Creates a `.git` directory
- Working directory vs Repository

The Three States of Files



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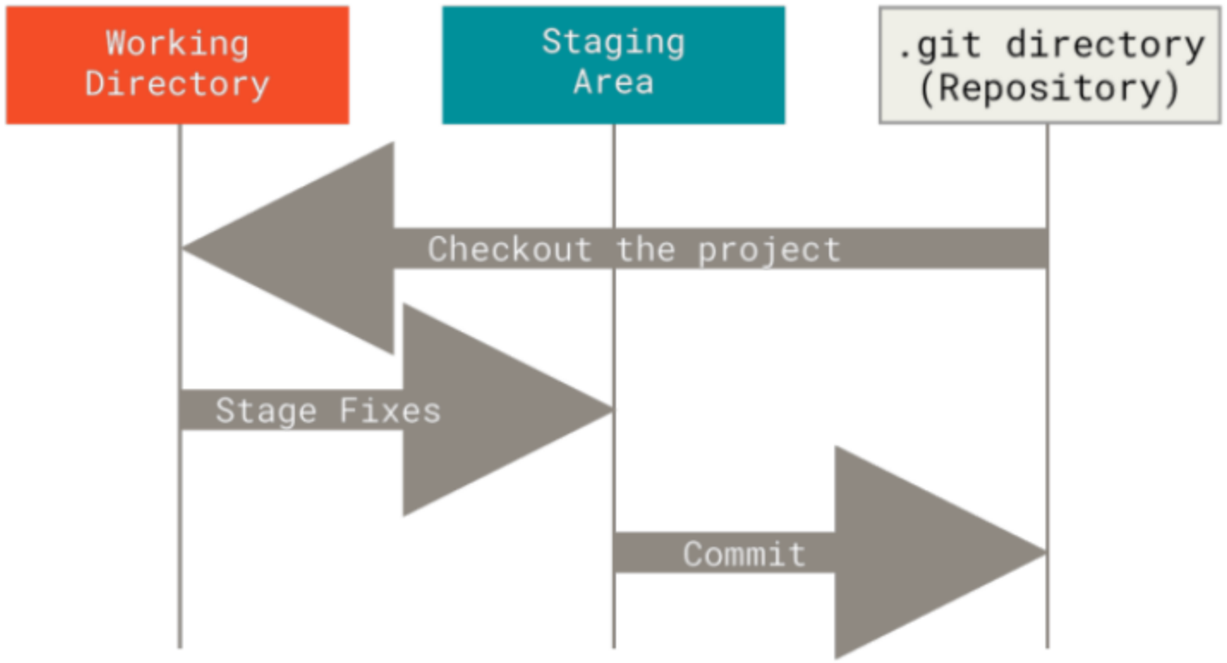

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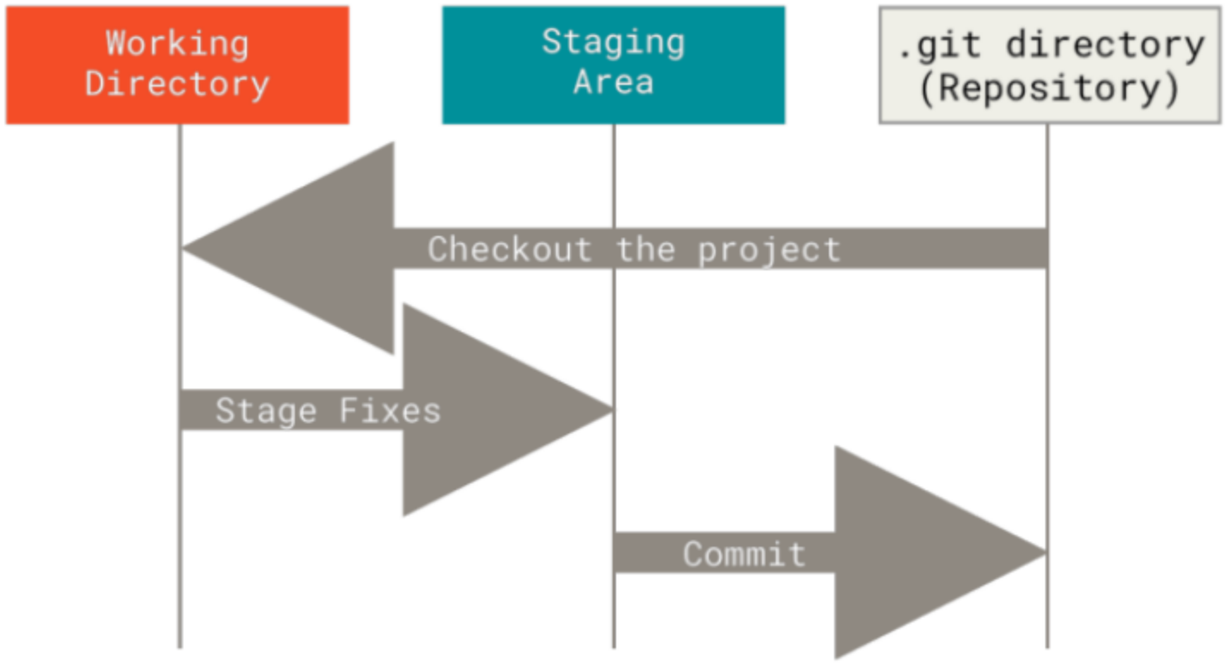

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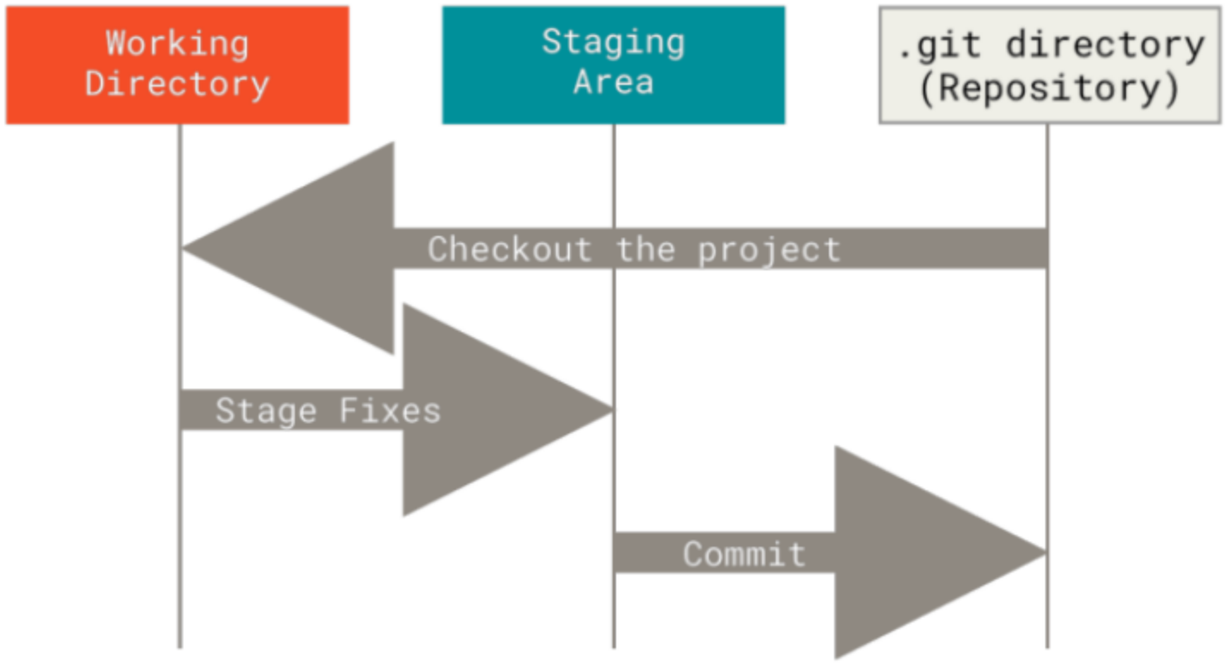

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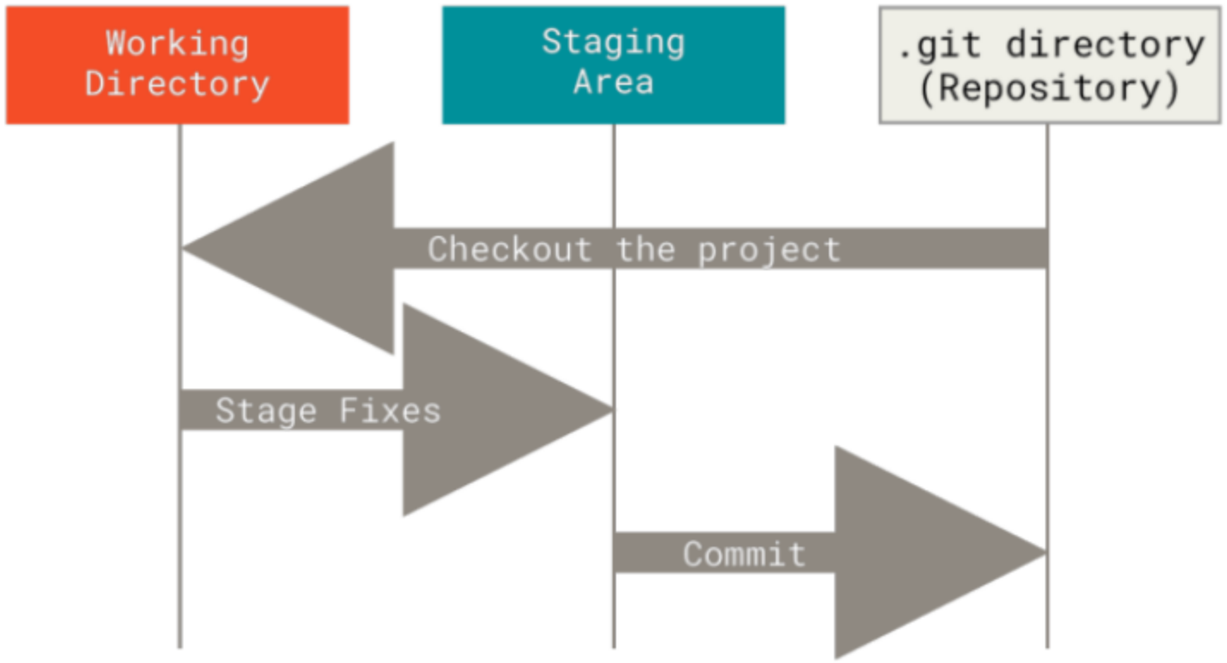

- Modified
- Staged
- Committed

Staging Area = deliberate commit preparation

First Commit

Commands

```
echo "Version 1" > notes.txt
git status
git add notes.txt
git commit -m "Initial commit"
```

- Commit = snapshot
- Commit message explains intent
- Git maintains integrity using hashes

Modified but Unstaged

Commands

```
echo "Version 2" >> notes.txt
git status
git diff
```

After add: `git diff --staged`

Question:

Why does Git not automatically commit this change?

Partial Staging, Ignoring and Removing Files, Logs

Commands

```
git add -p
```

- Stage changes interactively
- Build logical commits

Ignore files: Specify patterns in `.gitignore`

Remove files: `git rm [--cached] file`

Moving files: `git mv old new`

Explore: `git log`, `git log -p`

Undoing

Commands

```
git commit -m 'Initial commit'
git add forgotten_file
git commit --amend

git restore --staged file    # for unstaging

git restore file   # to discard changes
```

Access Previous Versions

```
git checkout <commit-hash>
git checkout <tag>

git restore --source=HEAD~1  file.txt

git show  <commit-hash>:</path/to/file>
git show HEAD~1:</path/to/file>
git diff <commit-hash> <commit-hash> -- /path/to/file
```

A new command `git switch` is recommended in place of `checkout`.